

Gabriel Pedde Ungureanu | Physics PhD & ML Scientist

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Profile

PhD physicist (6 papers) with hands-on AI/ML experience at EuroHPC scale: RLHF fine-tuning of Llama-3.2 (+11 pp on GSM8K), multi-agent LLM systems, and distributed GPU training on Leonardo (CINECA). Strong mathematical grounding in QFT translates directly to probabilistic modelling and rigorous ML research.

Professional Experience

SISSA

Doctoral Researcher

Trieste, Italy

2022–2026

Verified infrared duality conjectures in 3d $N=2$ supersymmetric QFT (mirror symmetry, particle-vortex dualities) by computing and matching superconformal indices and S^3 partition functions via symbolic computation on Ulysses (SISSA EuroHPC). First-year curriculum: advanced QFT, string theory, SUSY, AdS/CFT; deep foundations in representation theory and complex analysis.

Education

SISSA / ICTP Joint Programme

Master in High Performance Computing and AI (MHPC)

Trieste, Italy

2025–2026

Machine learning, deep learning, reinforcement learning, LLM agents, and distributed GPU computing. Hands-on work on **Leonardo** (CINECA), Ulysses (SISSA), Argo (ICTP).

Università Cattolica del Sacro Cuore

BSc Mathematics & MSc Physics

Brescia, Italy

2017–2022

Theoretical physics, QFT, condensed matter; strong analytical foundations.

Awards & Publications

Best graduate of the year, MSc Physics — Università Cattolica del Sacro Cuore

110/110 *cum laude* — MSc Physics | 110/110 *cum laude* — BSc Mathematics

Istituto Toniolo merit fellowship — BSc, Università Cattolica del Sacro Cuore

6 peer-reviewed papers in theoretical physics — [INSPIRE-HEP](#) | [Google Scholar](#)

Selected Projects — AI & ML

LLM RL fine-tuning: Implemented RLOO/PPO from scratch to fine-tune **Llama-3.2-1B** on GSM8K with a rule-based reward signal and KL regularisation; **+11 pp answer accuracy** (35 % $\rightarrow$ 46 %), format compliance from 51 % to 100 %. Distributed on **Leonardo** (CINECA). Stack: PyTorch, TRL, SLURM.

Multi-Agent LLM System: Opinion-dynamics experiments: $N = 4$ Qwen2.5 agents, **5 model sizes** (0.5B–14B), 10 rounds. Models $\geq 3B$ converged to $< 10^{-3}$ **error** at round 0 — a sharp transition driven by raw capacity, not training rounds. Stack: Ollama, Python.

Distributed GPU: CUDA kernel development through to cuBLAS; distributed dense linear algebra with **MPI + CUDA**. Lattice Boltzmann on a 2048×2048 grid: $\sim 120 \times$ **GPU speedup** over CPU (0.75 s vs 91.5 s, 806 GB/s bandwidth). MPI + OpenMP atmospheric simulation: near-linear scaling $2 \rightarrow 32$ nodes (3087 s $\rightarrow$ 317 s).

Technical Skills

AI / ML: PyTorch, Hugging Face Transformers & TRL, scikit-learn, cuML

LLM / Agents: RLHF, REINFORCE/RLOO, PPO, GRPO, fine-tuning, RAG, MCP, CrewAI, Ollama

ML methods: CNNs, autoencoders, representation learning, dimensionality reduction, Bayesian inference, clustering

Programming: Python, C++, C, Fortran, Mathematica, $\LaTeX$

Scale / HPC: CUDA, cuBLAS, OpenACC, MPI, OpenMP, SLURM, PETSc/SLEPc

MLOps: Docker, Kubernetes, Singularity, AWS, Git, HDF5/NetCDF

Clusters: Leonardo (CINECA), Ulysses (SISSA), Argo (ICTP)

Languages: Italian (native) | English (C1) | Romanian (A2)

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